

Chapter 10: Cell and Cell Organelles

Textbook Exercise Solutions & Answers



1. Who am I?

(a) I am ATP producing factory.

Mitochondria

(b) I am single-layered but maintain cellular osmotic pressure.

Vacuole

(c) I support the cell, but I am not cell wall. I have a body resembling net.

Endoplasmic reticulum

(d) I am chemical factory of the cell.

Chloroplasts in case of plant cells can be called chemical factory as they synthesise carbohydrates. Ribosomes also synthesise proteins, so those can also be called chemical factory. Golgi complex is secretory in function, hence it can be also called factory. Mitochondria though mainly referred to as powerhouse of the cell, it is also mentioned as chemical factory by some authors.

(e) Leaves are green because of me.

Chloroplast.

2. What would have happened? If.....

(a) RBCs had mitochondria.

Mitochondria continuously carry out oxidation and form energy inside the cell. They produce energy-rich compound, ATP. In this process, they utilize carbohydrates, fats and proteins present in the cell. If RBCs has mitochondria, they would have used oxygen for this purpose than carrying it to all the cells of the body. The cells would not have obtained oxygen.

(b) There had been no difference between mitochondria and plastids.

Mitochondria carry out oxidation of carbohydrates, fats, etc. with the help of enzymes. Plastids are synthesising carbohydrates with the help of solar energy and chlorophyll. Both the cell organelles have their own sets of different enzymes as per their role. If there would have been no difference between mitochondria and plastids, the specific functions would not have been taken place.

(c) Genes had been absent on the chromosomes.

Genes are functional segments on the chromosomes which are responsible for transmitting the hereditary information.

(d) Plasma membrane had not been selectively permeable.

Selectively permeable membrane allows some substances to enter the cell, while prevents other unwanted or harmful substances. If plasma membrane would not have been selectively permeable, there would be no control over entry and exit of any substances. The process of osmosis would also be erroneous in such case.

(e) Plants lacked anthocyanin.

If plants lacked anthocyanin, no part of the plant would display purple or blue colour. Anthocyanin attracts the insects for pollination and seed dispersal. These processes will be affected due to lack of anthocyanin. These pigments are also said to be protective in nature for the plant. This protection will not be given to the plant in absence of anthocyanin.

3. Who is odd man among us? Give reason.

(a) Nucleolus, mitochondria, plastids, endoplasmic reticulum

Nucleolus. (All the others are cell organelles but nucleolus is not a cell organelle present in cytoplasm.)

(b) DNA, Ribosomes, Chlorophyll

Chlorophyll. (DNA and Ribosomes are present in plant as well as in animal cells. Chlorophyll is present only in plant cells.)

4. Give functions.

(a) Plasma membrane.

1. Plasma membrane acts as a selectively permeable membrane. It allows entry of those useful substances which are needed for the cell. It does not allow entry of the harmful and unwanted substances.
2. Plasma membrane keeps the homeostasis in the cell. The cell is kept in steady state even if the external environment changes.
3. Plasma membrane is responsible for processes of endocytosis and exocytosis.
4. The processes of diffusion and osmosis are possible only due to plasma membrane.
5. In animal cells, plasma membrane is the outermost protective covering of the cell.

(b) Cytoplasm.

1. All the cell organelles are spread in the cytoplasm of a cell.
2. The cytoplasm is the medium for many cellular chemical reactions.
3. The cytosol which is the part of cytoplasm other than cell organelles stores many vital substances like amino acids, glucose, vitamins, etc.
4. Cytosol also helps in the cellular movements.

(c) Lysosome.

1. Lysosome helps in the destruction of attacking viruses and bacteria and thereby help in the immune response.
2. Lysosomes act as demolition squads. They destroy worn-out cellular organelles and organic debris. This process is called autolysis which All Pagesmeans self-destruction.
3. They are also called suicide bags as in a worn out, damaged or old cell, lysosomes automatically burst. The lytic enzymes present in the lysosome digest their own cells.
4. Lysosomes can digest stored proteins, fats during starvation.

(d) Vacuole.

1. Vacuoles maintain the osmotic pressure of the cell.
2. Various metabolic byproducts and end products such as glycogen, proteins, water, etc. are stored in the lysosome.
3. In food vacuole of amoeba, the food is temporarily stored till digestion. In other animal cells, vacuoles can store waste products and food.
4. Vacuoles of plant cells can provide turgidity and rigidity as it contains good amount of cell sap.

(e) Nucleus.

1. Nucleus is the controlling centre for the entire cell.
2. It controls all metabolic activities of the cell.
3. The cell division is possible due to the nucleus.
4. The chromosomes present in the nucleus carry the genes. These genes are responsible for the transmission of hereditary characters from parental generation to the next generations.

5. Who gives me the colour? (Select the correct option).

Column 'A'	Column 'B'
1. Red tomato	a. Chlorophyll
2. Green leaf	b. Carotene
3. Carrot	c. Anthocyanin
4. Violet	d. Lycopene

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